Multimodal 3D Perception BEV Occupancy Prediction

RGB + LiDAR + IMU Sensor Fusion on nuScenes-mini

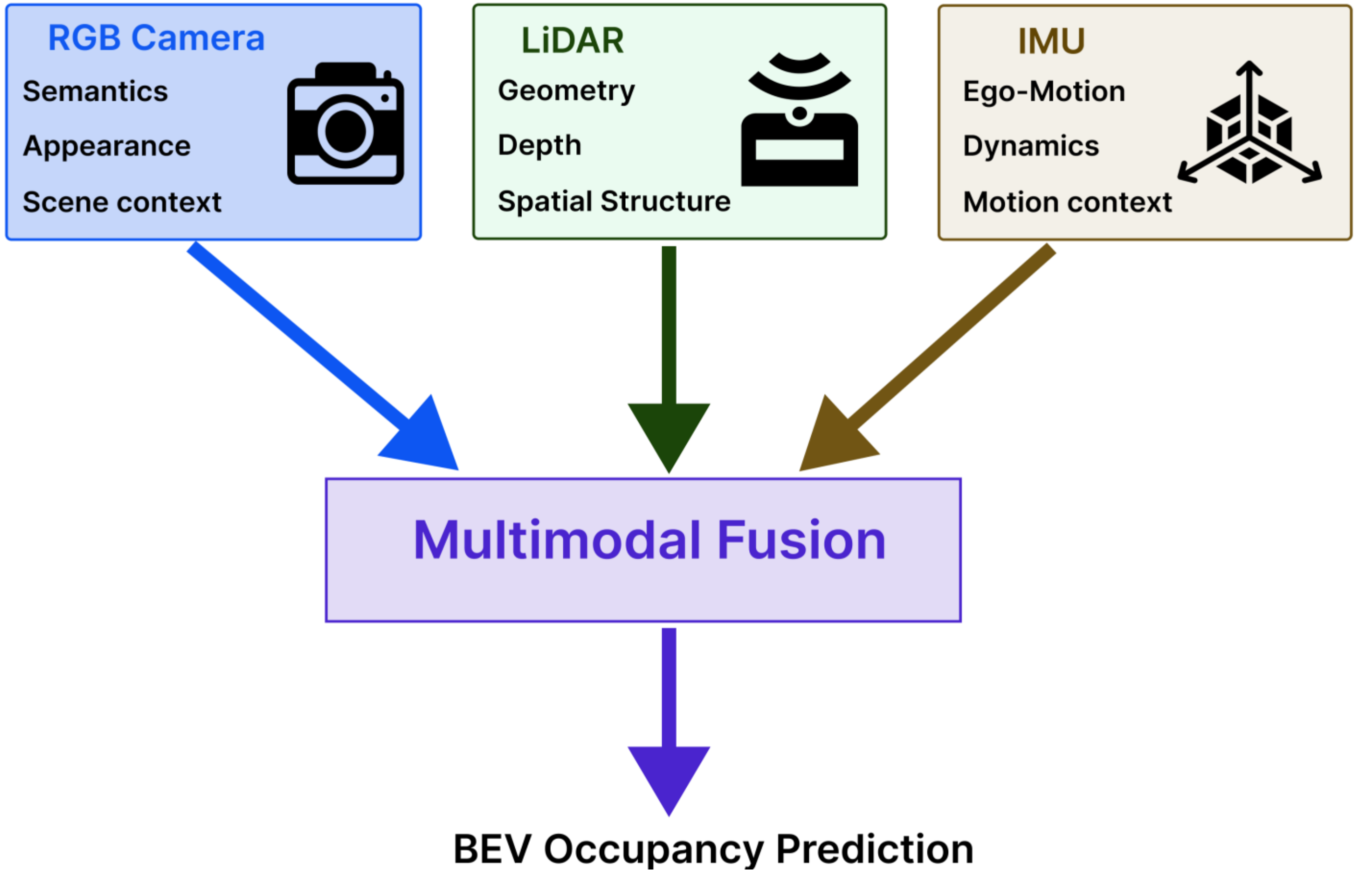
Johnson Opadere

GitHub Repository: https://github.com/Johnson-Opadere/multimodal-bev-occupancy

August 2026

Problem & Motivation

Why Multimodal BEV Perception?



Goal: Combine complementary semantic, geometric, and motion cues for accurate, robust, deployment-ready BEV perception.

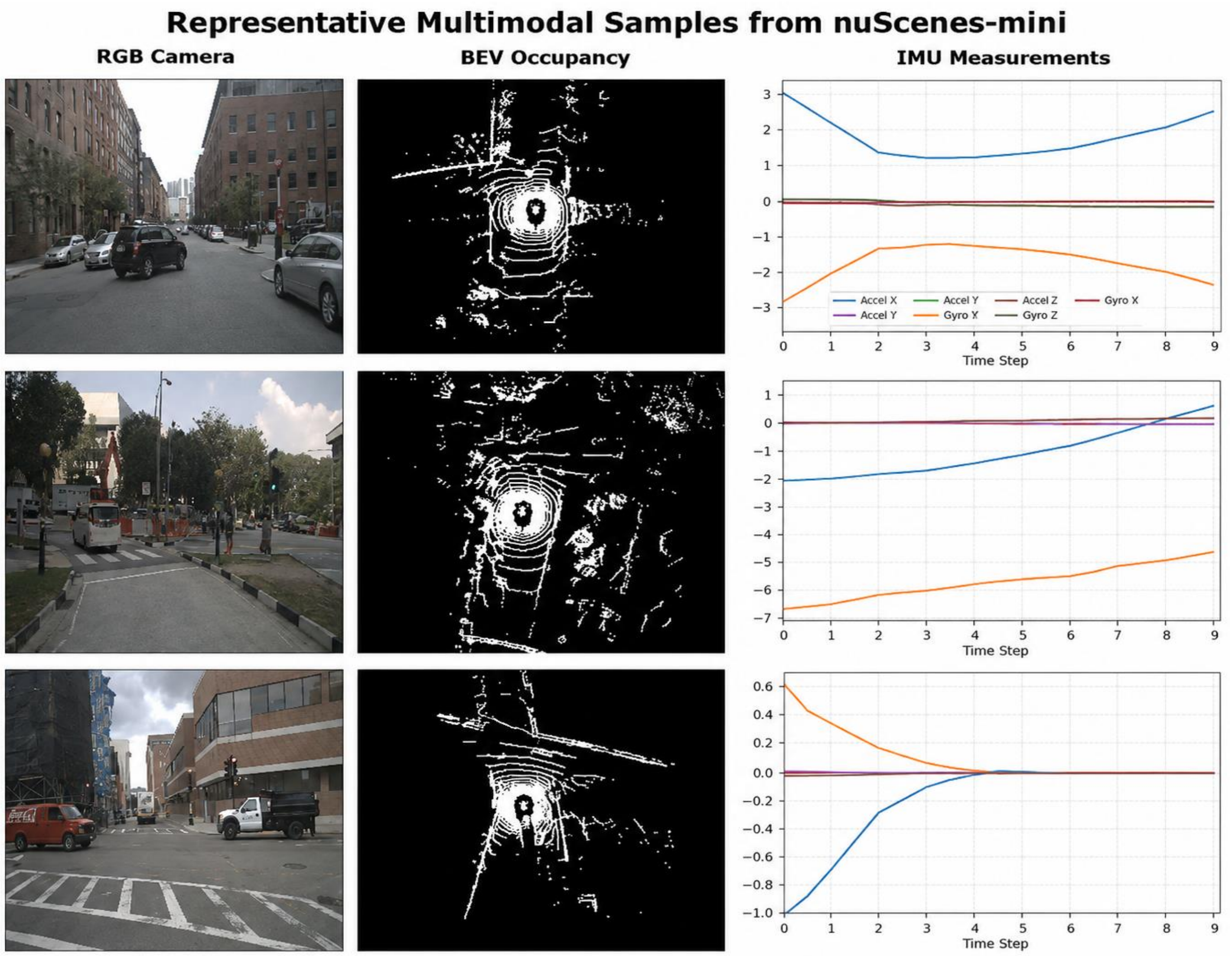
Dataset & Multimodal Inputs

Synchronized Multimodal Inputs

Dataset: nuScenes-mini

Task: Dense binary BEV occupancy prediction

Modalities: RGB camera + LiDAR + IMU



RGB provides appearance and semantic context, LiDAR provides spatial geometry, and IMU provides ego-motion context.

System Architecture

Multimodal BEV Perception Architecture

- Independent RGB, LiDAR, and IMU encoders
- Modality-specific latent representations
- Multimodal feature fusion
- Decoder + 1x1 prediction head
  → dense BEV occupancy

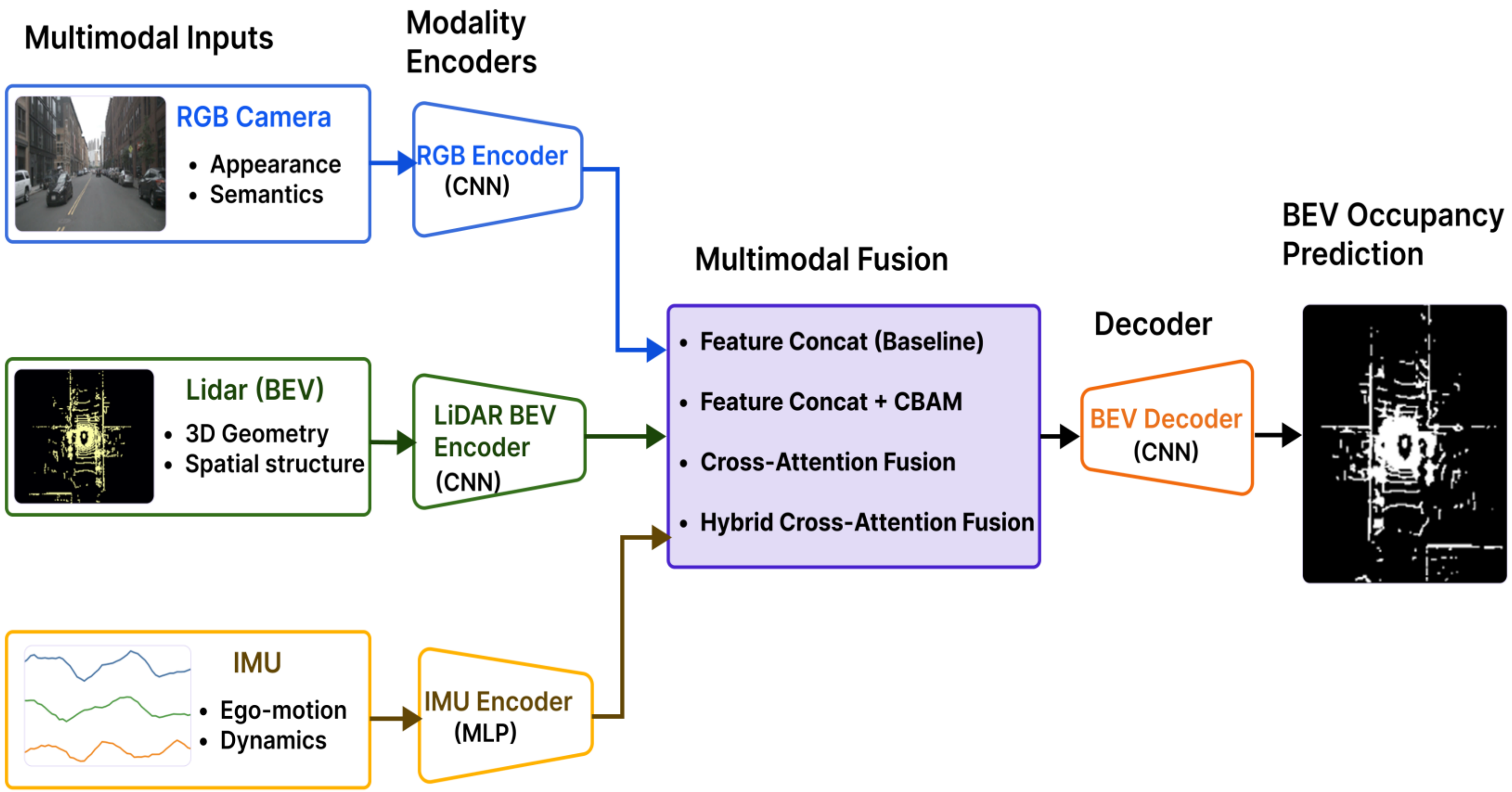
$$\hat{Y}_{BEV} = D(F(E_{RGB}(x) + E_{LiDAR}(l), E_{IMU}(m)))$$

$E_{\cdot}$: Modality-specific encoder

F : Fusion operation

D : BEV decoder/prediction network

Independent modality encoders learn complementary features that are fused to generate dense BEV occupancy predictions.



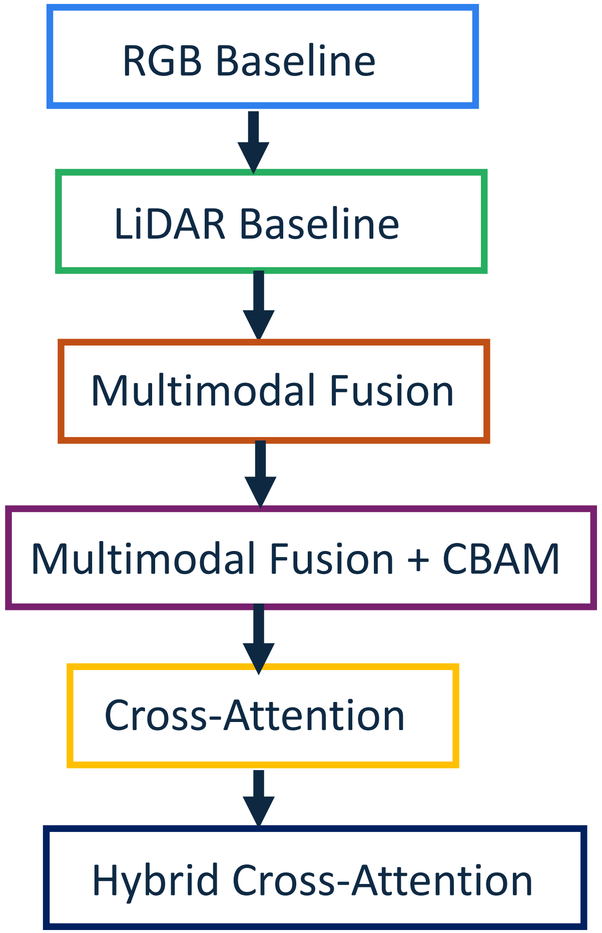
Architecture Evolution

Progressive Architecture Development Evolution

Mode	Purpose
RGB Baseline	Visual feature learning
LiDAR Baseline	Geometric feature learning
Multimodal Fusion	Feature concatenation
Multimodal Fusion + CBAM	Channel & spatial attention
Cross Attention	Cross-modal interaction
Hybrid Cross-Attention	Attention + residual feature preservation

Progressive architectural refinement consistently improved multimodal feature integration and overall BEV perception performance.

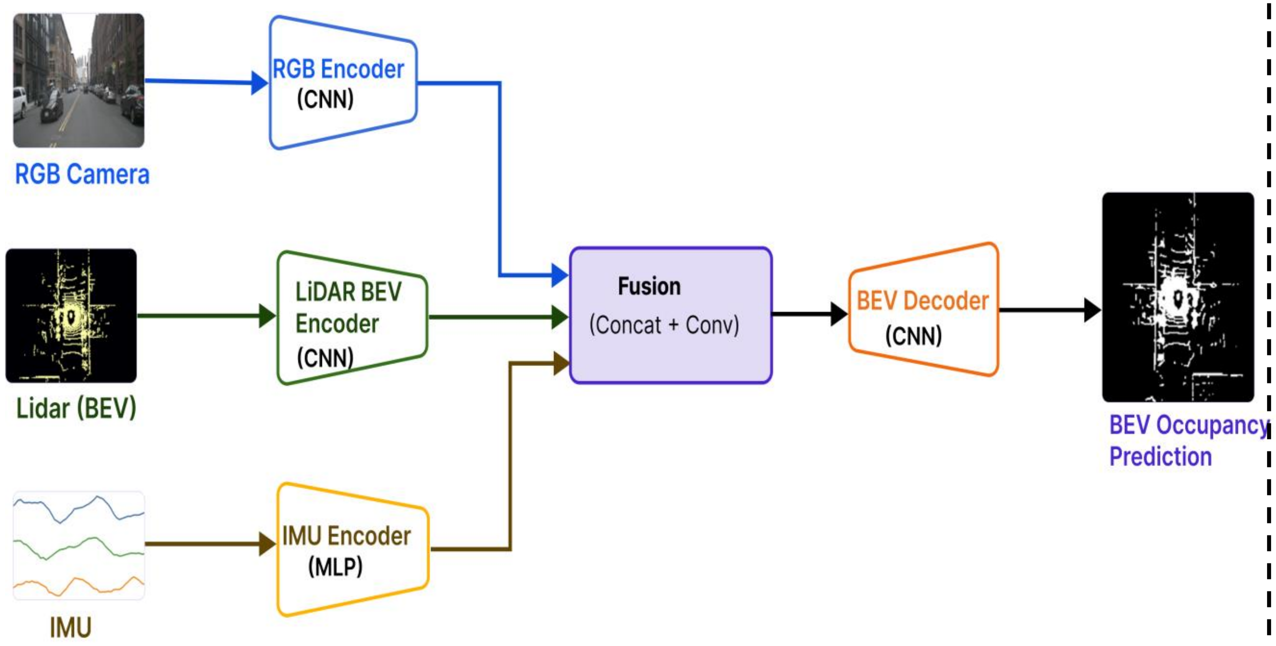
Architecture Evolution



From Feature Fusion to Attention Refinement

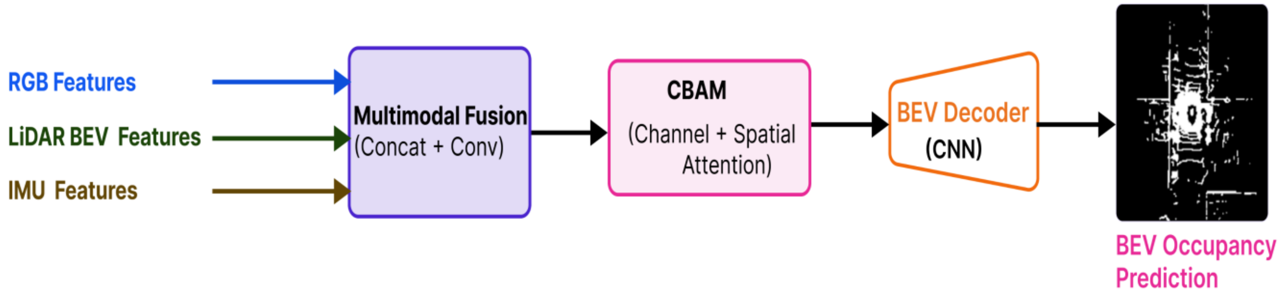
Baseline Fusion Architectures

Fusion Baseline



- Concatenate
- Convolution

Fusion + CBAM



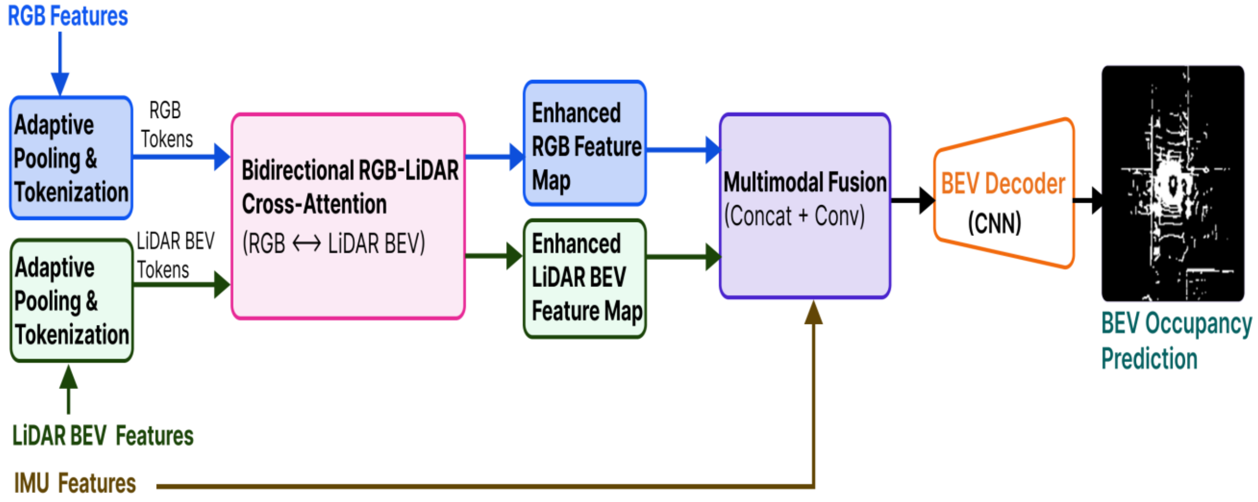
- Adaptive feature weighting
- Improved feature selection

CBAM extends baseline feature fusion with channel and spatial attention for adaptive feature refinement.

From Cross-Attention to Hybrid Fusion

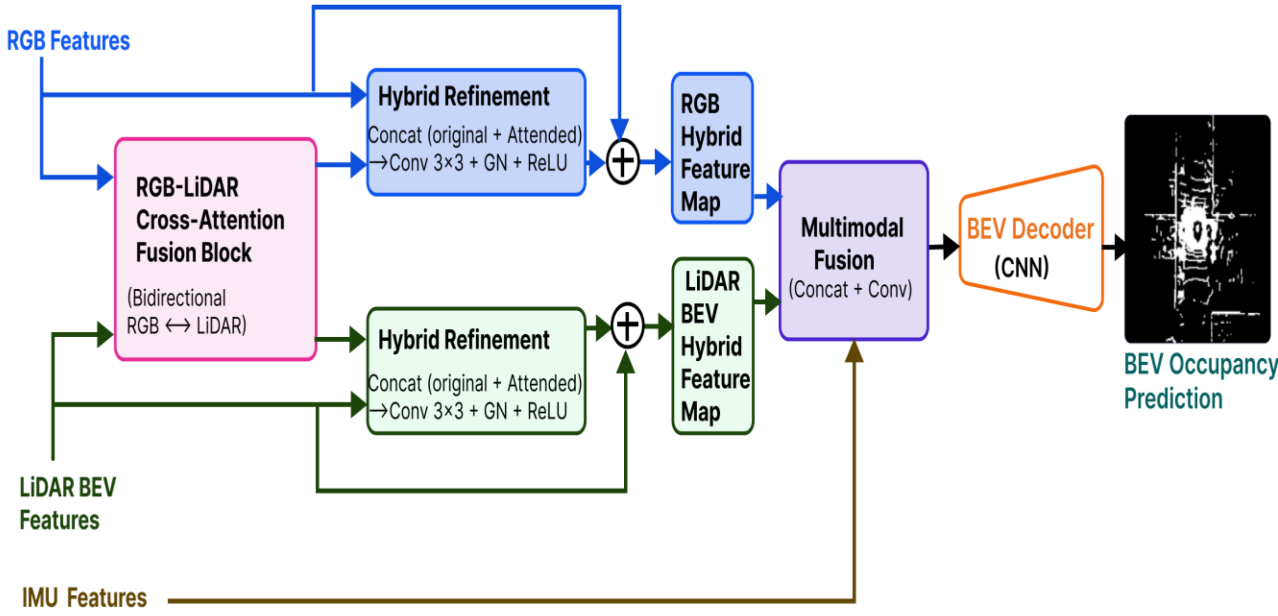
Attention-Based Fusion

Cross-Attention



- Attention
- Learned interaction

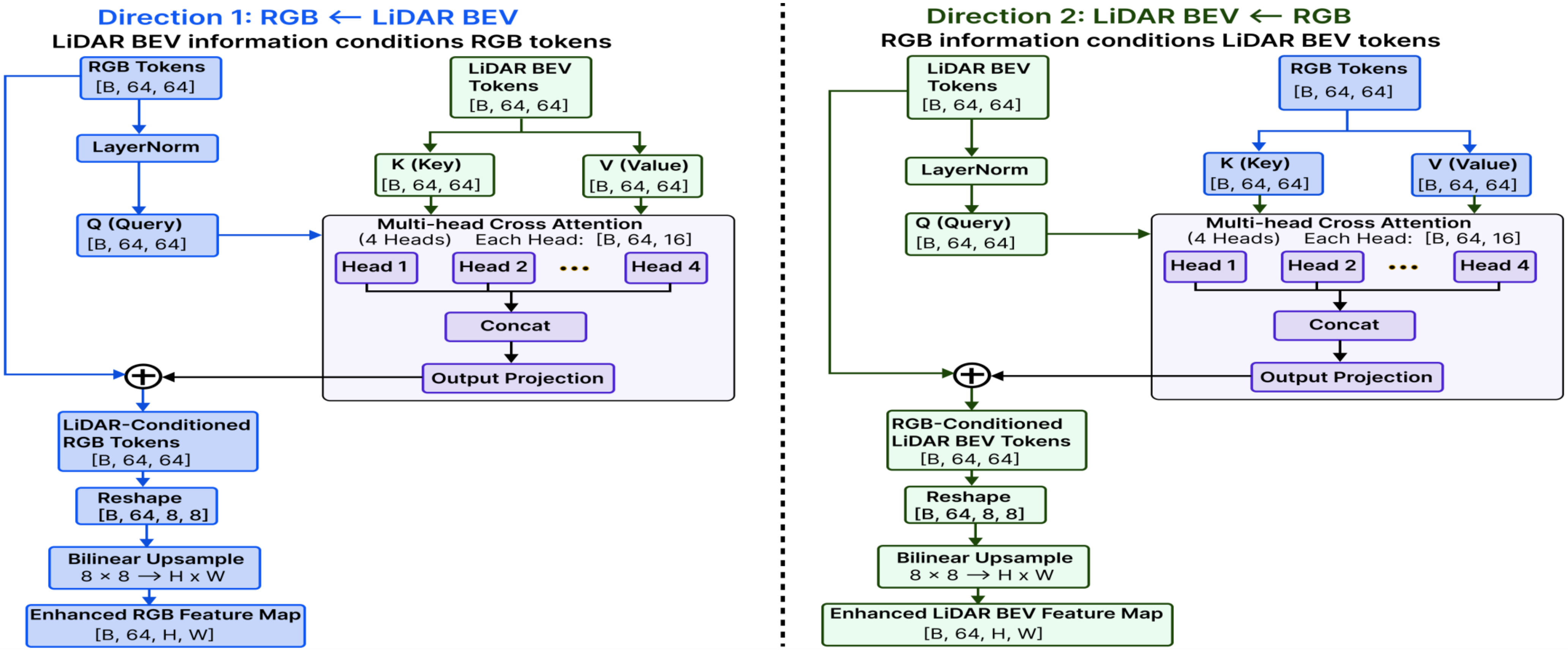
Hybrid Cross-Attention



- Attention
- Residual preservation

Hybrid Cross-Attention preserves modality-specific features while learning explicit cross-modal interactions.

Bidirectional RGB-LiDAR Cross-Attention



RGB ← LiDAR BEV

Geometry guides semantic feature refinement

LiDAR BEV ← RGB

Appearance guides geometric feature refinement

Bidirectional attention enables explicit RGB–LiDAR interaction beyond feature concatenation.

$$Attention(Q,K,V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)$$

Quantitative Results

Best Overall:

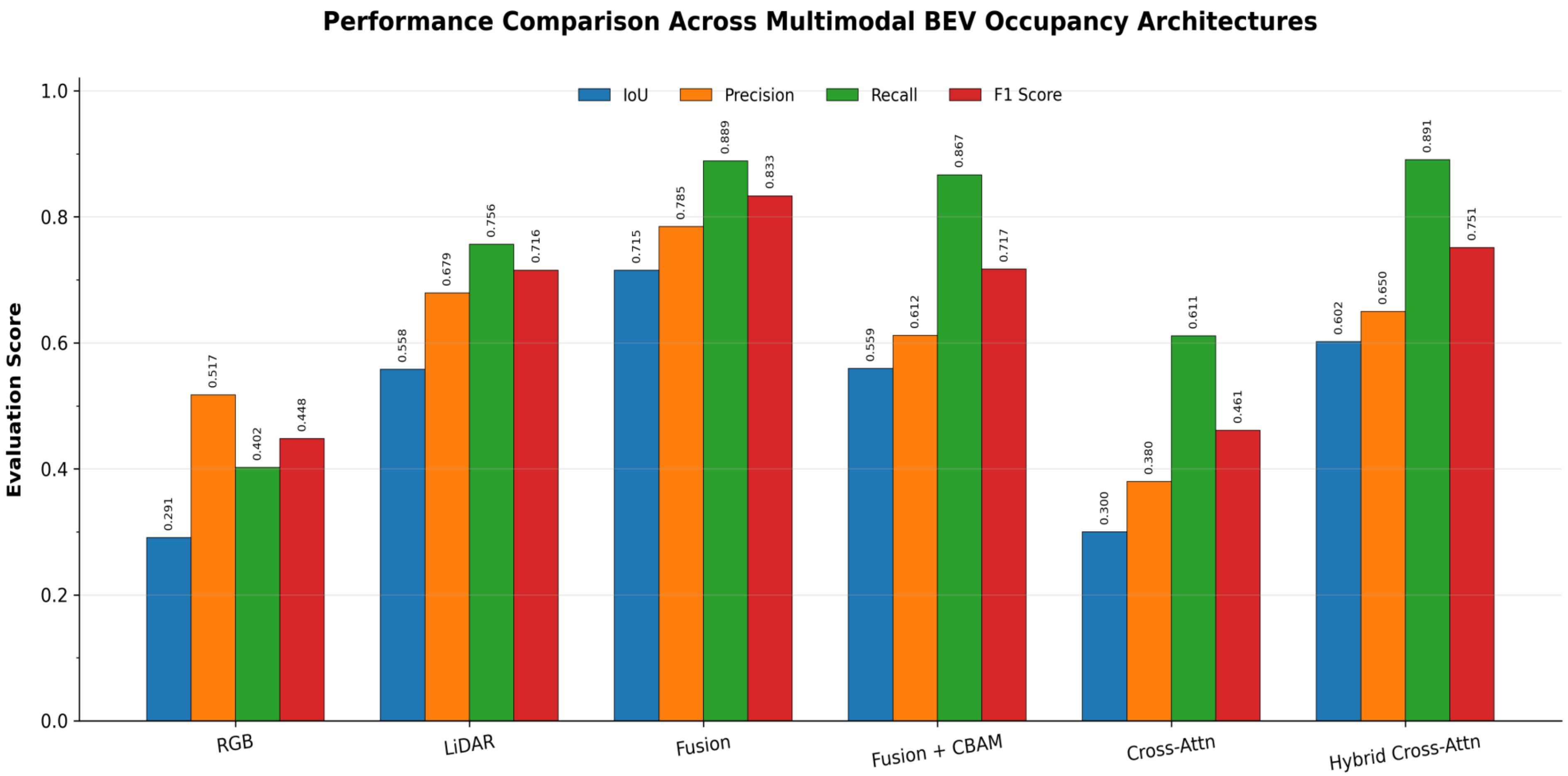
Multimodal Fusion

IoU:	0.7150
Precision:	0.7847
F1:	0.8333

Highest Recall:

Hybrid Cross-Attention

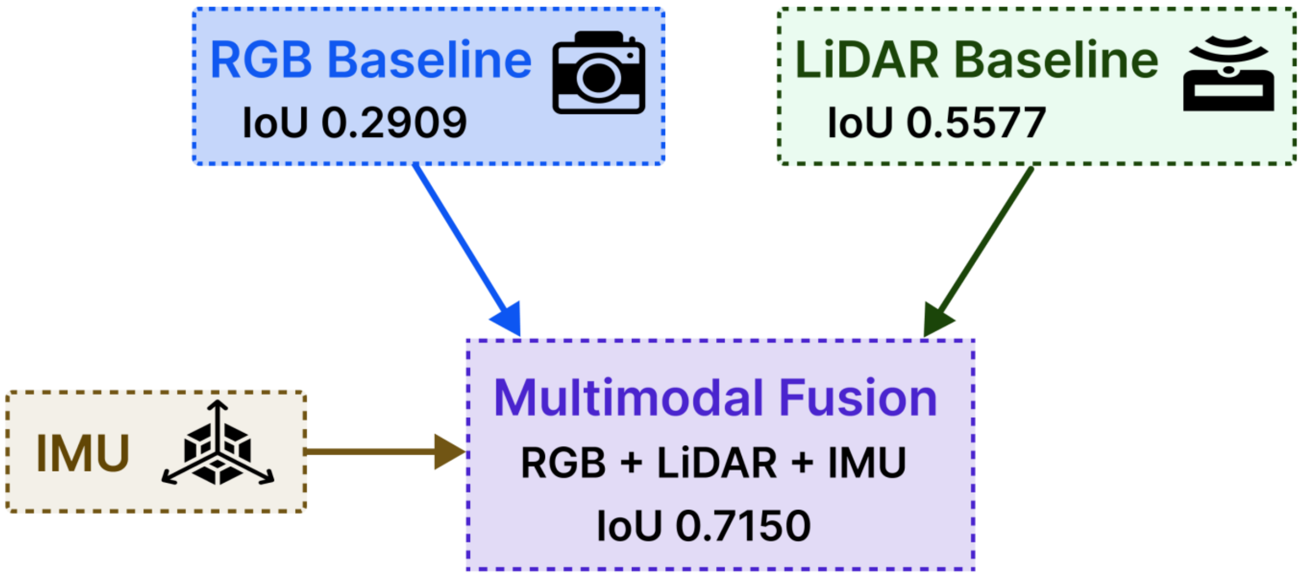
Recall:	0.8906



Key result: Multimodal Fusion achieved the strongest overall performance, while Hybrid Cross-Attention achieved the highest recall.

What Actually Drove Performance?

Sensor Contribution



Complementary multimodal sensing substantially improved IoU over either unimodal baseline.

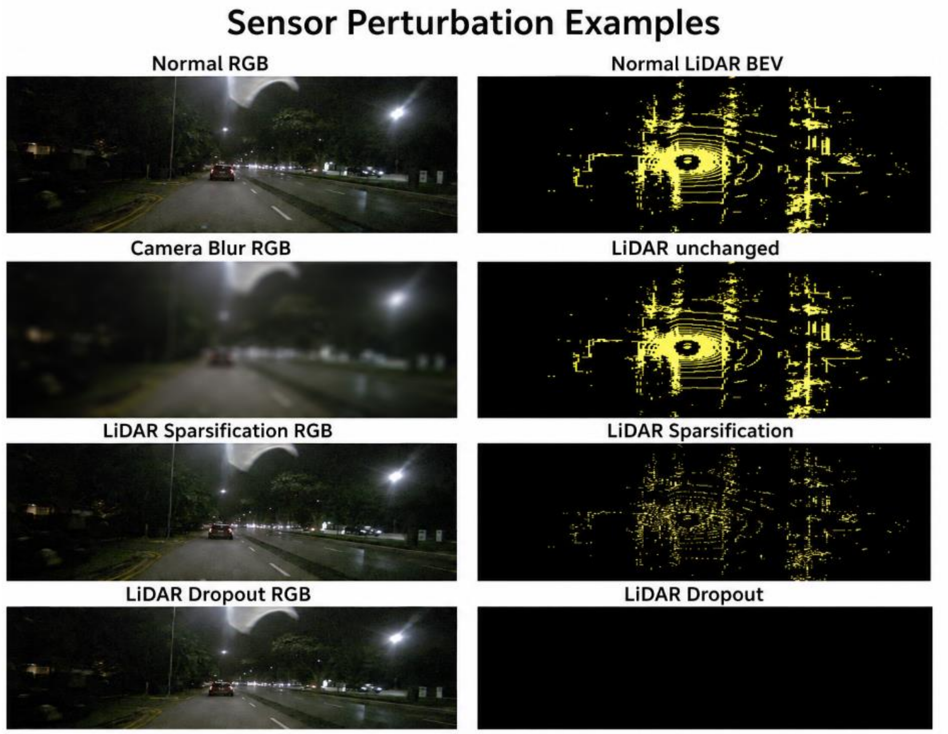
Modality diversity mattered more than fusion complexity; simple multimodal fusion delivered the strongest overall IoU.

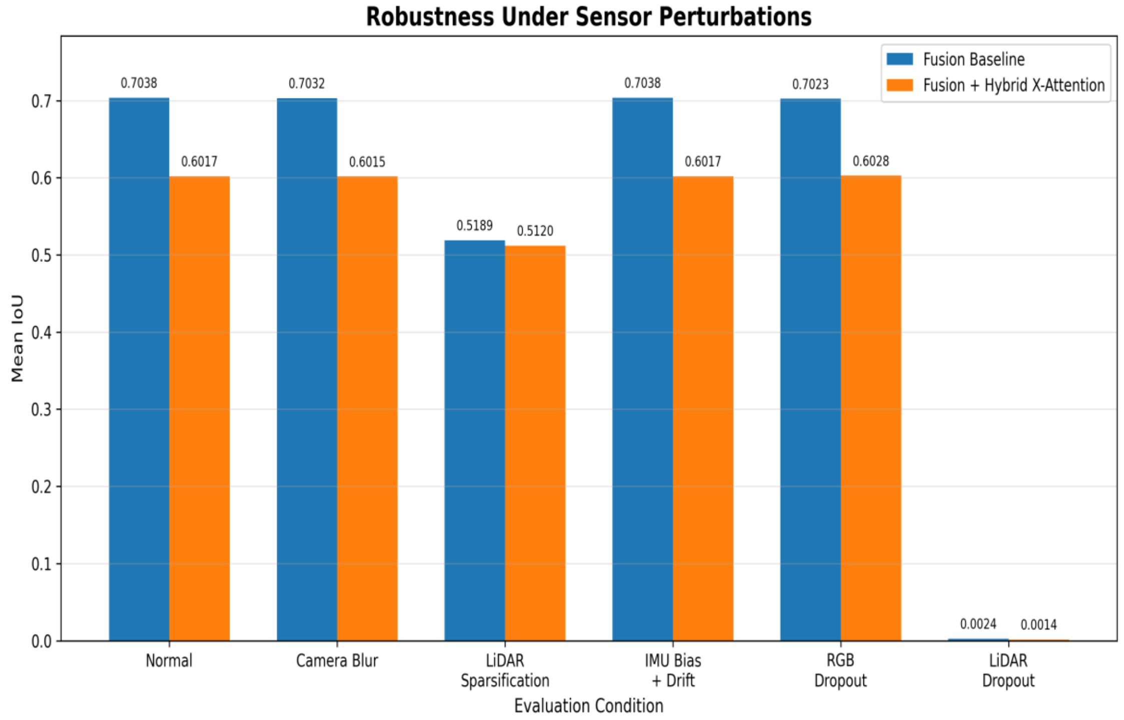
Fusion Complexity

Architecture	IoU
Multimodal Fusion	0.7150
Multimodal Fusion + CBAM	0.5592
Cross Attention	0.3000
Hybrid Cross-Attention	0.6017

More sophisticated fusion did not guarantee better generalization.

Robustness Under Sensor Degradation



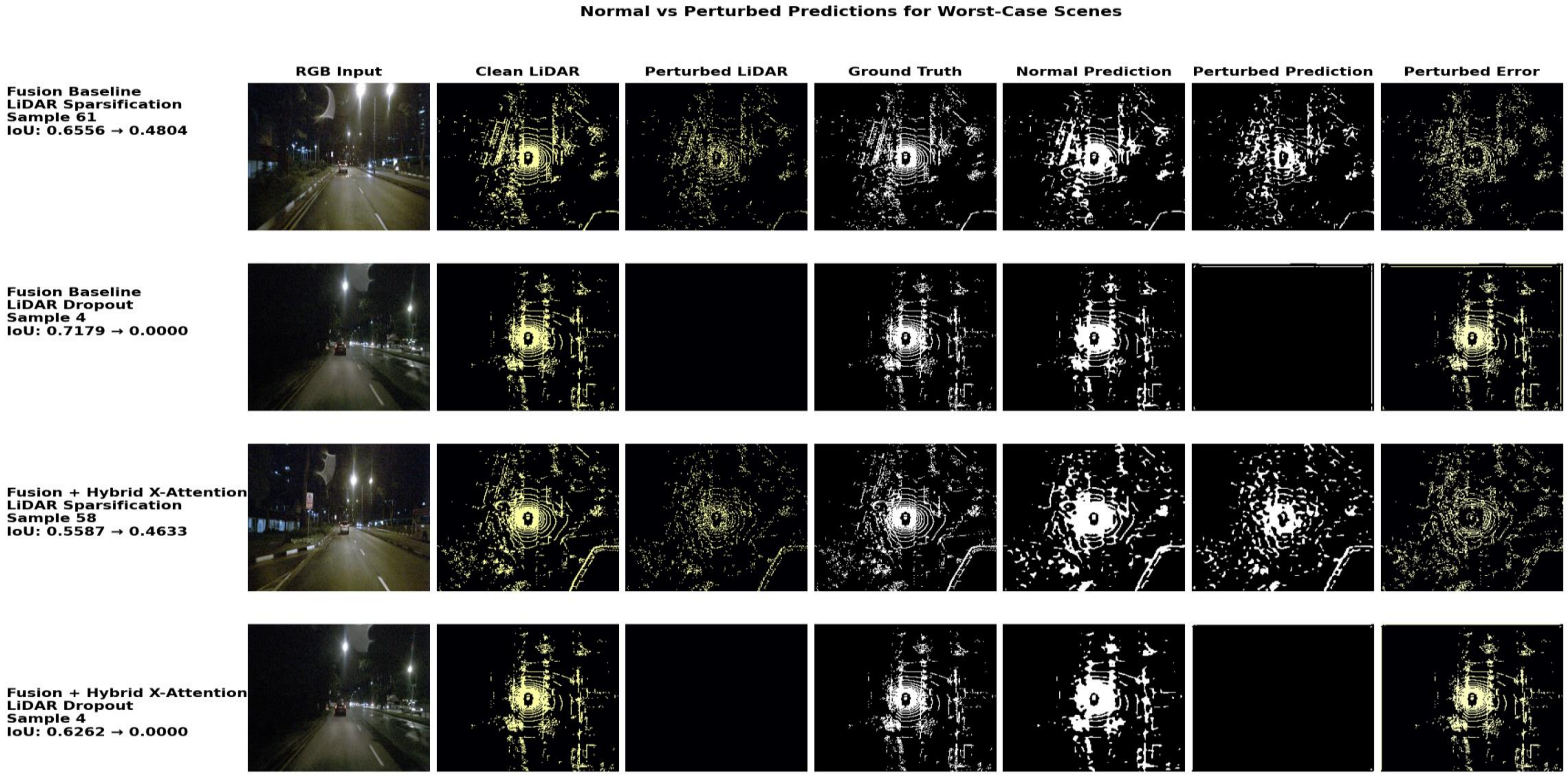


Key Findings

Perturbation	Performance Impact
Camera blur	Negligible degradation
IMU bias/drift	Minimal degradation
LiDAR sparsification	Fusion: −26.28%; Hybrid Cross-Attention: −14.92%
LiDAR dropout	Near total performance collapse

Critical dependency**:** BEV occupancy performance is strongly dependent on LiDAR geometry.

Failure Analysis: LiDAR as the critical modality



Sensor Condition	Observed Effect
Normal sensing	Stable Bev occupancy
LiDAR degradation	Reduced geometric structure
LiDAR dropout	Prediction collapse

Failure mode: Multimodal fusion does not guarantee modality redundancy; the system remains critically dependent on LiDAR geometry.

Deployment & Engineering

Best Model

Multimodal Fusion

IoU: 0.7150

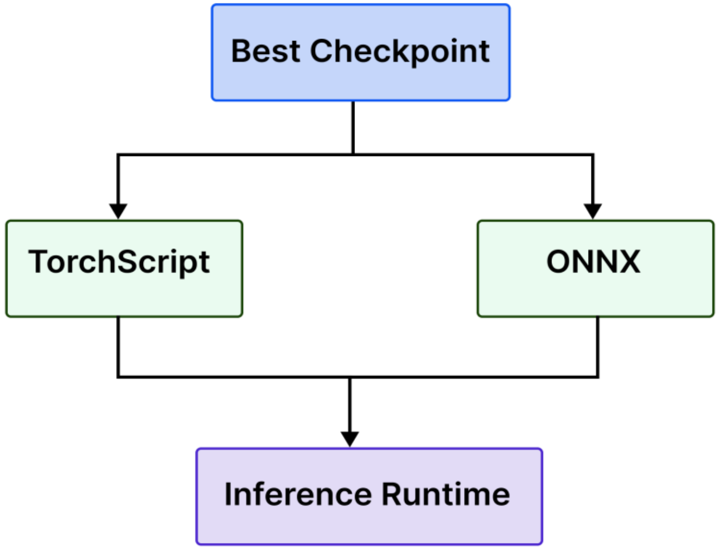
GPU Latency

2.56ms; ~ 390 FPS

Model Size

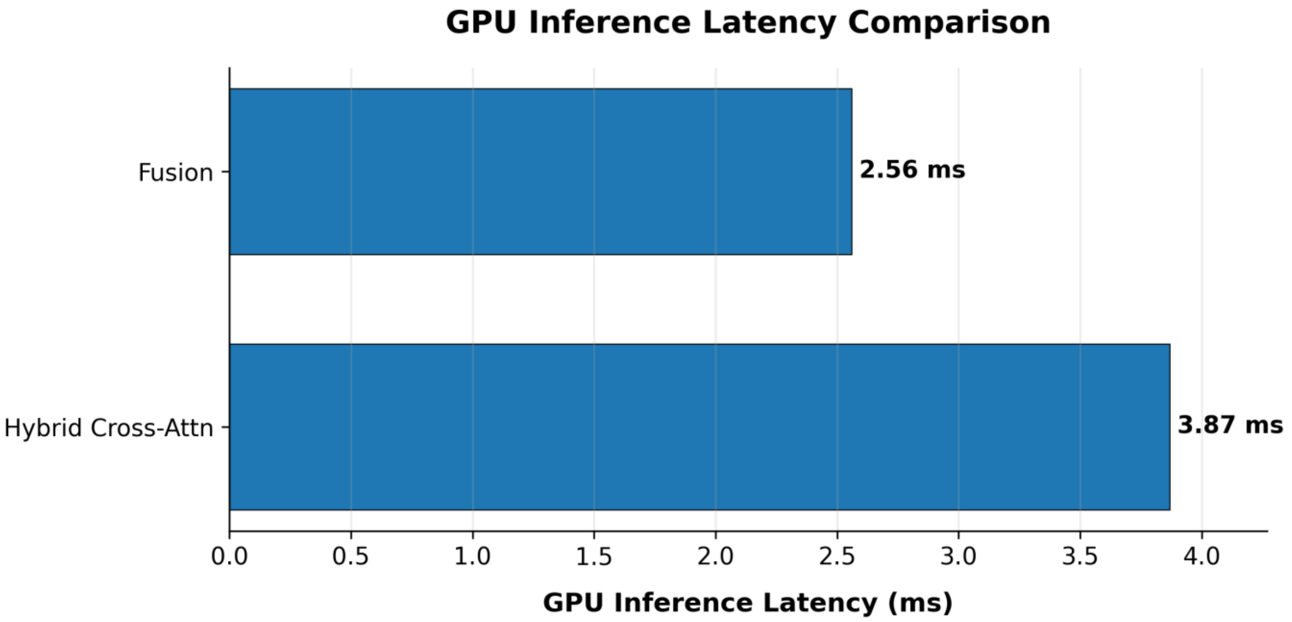
1.39 MB; 364,913 params

Deployment path



Multimodal Fusion provided the best accuracy–latency–memory trade-off.

Latency Benchmark



Key Findings & Future Work

Key Findings

1. **Multimodality Works**

   RGB + LiDAR + IMU
   IoU: 0.7150

2. **Complexity ≠ Performance**

   Simple fusion outperform attention variants.

3. **LiDAR is the critical sensor**

   Dropout caused near-total performance collapse.

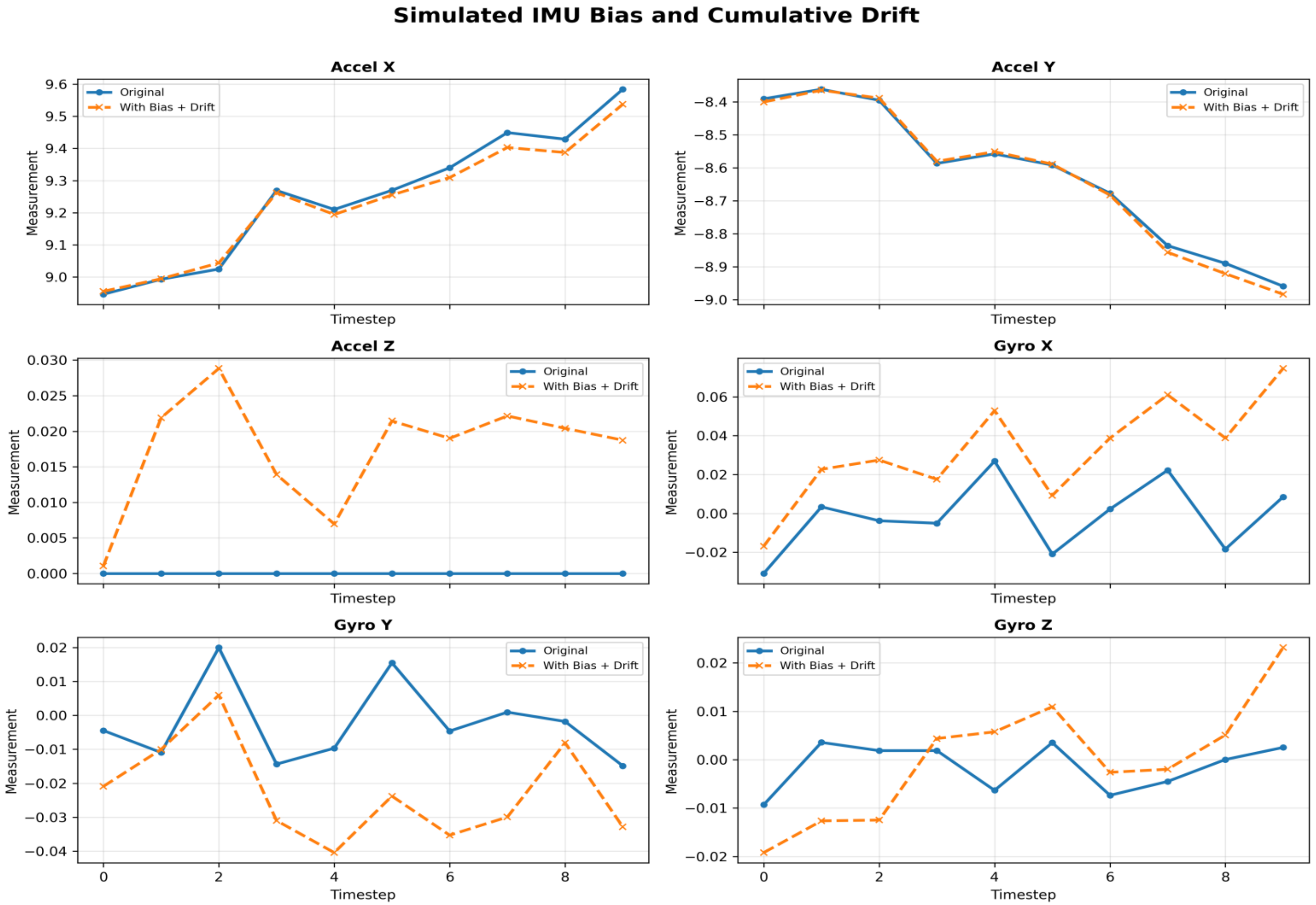
4. **Deployment is practical**
   - 1.39 MB model size
   - 2.56ms GPU latency

Future Work

- Temporal sensor fusion across multiple frames
- Full nuScenes benchmark
- Modern BEV architectures
  BEVFormer / BEVFusion / BEVDet
- Deployment optimization
  TensorRT / FP 16 / INT8

Project 3 demonstrates end-to-end multimodal perception engineering—from sensor fusion and architectural experimentation to robustness analysis and deployment.

Appendix



The tested IMU bias and drift produced minimal degradation, indicating limited sensitivity to IMU perturbation in this setup.

Test Setup

Synthetic bias and temporal drift applied to IMU signals

Result

Minimal impact on BEV performance